Marco Yang

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education

California Institute of Technology — On leave
B.S. Computer Science · GPA 3.91 · Pasadena, CA

skills

Languages	C, C++, Java, Go, Python, TypeScript
ML & AI	PyTorch, Agentic Frameworks, Computer Vision, Generative AI, Reinforcement Learning
Systems	CUDA, Linux, Git, Shell

experience

Member of Technical Staff · Mechanize — Sep 2026 – Present
- Evals and RL tasks.

Software Engineering Intern · Two Sigma — Jun 2026 – Aug 2026
- Related to debugging simulations.

Software Engineer Intern · Glean — Jun 2025 – Sep 2025
- AI Team at enterprise search and assistant startup.
- Built and deployed LLM-as-a-judge evaluations and automated evalset pipelines for agents.
- Shipped automated prompt optimization for Glean Agent Builder.
- Misc. features and bug fixes in Go agent backend.

Teaching Assistant · Intro to Software Design in C — Spring 2025
- Guided projects building a physics engine with collision and SDL graphics, and video game.

Summer Undergraduate Research Fellow · Caltech Vision (Perona) Lab — Jun 2024 – Aug 2024
- Cross-domain video action recognition with generative diffusion models.
- Extracted cross-attention maps from UNet in Stable Diffusion and Modelscope.
- Implemented LSTM, 3D CNN, Transformer classification heads on top of diffusion features.

projects

CUDA Qwen3.5 Inference — github
- CUDA autoregressive inference engine for Qwen3.5 with single-token prediction.
- Flash attention and Gated DeltaNet linear-attention kernels.

Catan RL Bot — github · play
- RL bot for Settlers of Catan trained with DAgger, PPO, and AlphaZero (WIP).
- ~70% win rate vs Catanatron baseline bots.

Caltech RAG — github · demo archived
- LLM RAG app as unified interface for Caltech course reviews and catalogs.
- Flask, LangChain, LangGraph, Qdrant.

relevant coursework

Data Structures, Algorithms, Machine Learning & Data Mining, Foundations of Machine Learning, Advanced Machine Learning, Large Language and Vision Models, GPU Programming, Computing Systems, Networks, Relational Databases, Applied Linear Algebra, Statistical Inference, Information Theory

olympiads

Putnam Top 1000 — Dec 2023
USACO Gold Division — Feb 2021
AIME Qualifier — 2019, 2020, 2021

fun facts

Caltech tennis team member. Formerly assistant program director for ACEing Autism (Caltech branch). Biked from Denver to NYC.